Project 6: Data Exploration and Visualization

Overview

In this project, you will apply data exploration and visualization techniques to a real-world dataset of your choice. You'll practice using pandas for data analysis and create meaningful visualizations to gain insights from your data. This project builds on your Python programming skills while introducing you to the exciting world of data science.

In Lab6, you explore the titanic dataset and you should use it as a sample and pick a dataset of your interest to do a similar data exploration and visualization.

Project Objectives

- Find and select a dataset from a real-world data source
- Formulate a clear inquiry question based on your chosen dataset
- Perform exploratory data analysis using pandas functions
- Create at least two meaningful data visualizations
- Analyze your findings and answer your inquiry question
- Reflect on the insights gained through data exploration

Data Sources

You can choose from various online data sources such as:

- Kaggle - Wide variety of datasets
- data.gov - U.S. government data
- Our World in Data - Global development data
- GitHub - Curated datasets
- Google Dataset Search - Search for datasets
- Or any other reliable data source you find interesting

Part 1: Dataset Selection and Initial Exploration

TODO Tasks:

1. Select Your Dataset: Choose a dataset that interests you. Make sure it has at least 100 rows and multiple columns with different data types (e.g., numerical, categorical, dates).

Please create either text or code cells in between the TODO tasks to do the work.

2. Import Libraries: Import the necessary libraries (pandas, matplotlib, seaborn) at the beginning of your notebook by creating code cells below this text cell.

```
!pip install pandas
!pip install seaborn
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/pytho
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/di
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: numpy!=1.24.0,>=1.20 in /usr/local/lib/python3
Requirement already satisfied: pandas>=1.2 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in /usr/local/lib/pyth
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/di
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-pac
```

3. Load Your Data: Load your chosen dataset using pandas and save it to a pandas dataframe. Please create a code cell below this text cell to do it.

4. Data Summary: Use the following functions to do inital exploration your dataset:

- info() - Display information about the dataset structure
- describe() - Show statistical summary of numerical columns
- head() - Display the first 5 rows (df.head())
- shape - Show the dimensions of the dataset (e.g. df.shape)
- columns - Get the column names (e.g. df.columns)

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You may use 5 code cells below this text cell to accomplish it.

```
import seaborn as sns
import pandas as pd

df = pd.read_csv('/student_performance.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 21 columns):
 #   Column                      Non-Null Count  Dtype  
---  ------                      --------------  -----  
 0   Name                        1000 non-null   object 
 1   Age                         1000 non-null   int64  
 2   Gender                      1000 non-null   object 
 3   Major                       1000 non-null   object 
 4   Living_Area                 1000 non-null   object 
 5   Parent_Education            1000 non-null   object 
 6   Scholarship_Status          1000 non-null   object 
 7   Part_Time_Job               1000 non-null   object 
 8   Extracurricular_Activities  1000 non-null   object 
 9   Internet_Access             1000 non-null   object 
 10  Study_Hours_Per_Day         1000 non-null   float64
 11  Sleep_Hours_Per_Day         1000 non-null   float64
 12  Attendance_Rate             1000 non-null   float64
 13  Stress_Level                1000 non-null   float64
 14  Exam_Proximity              1000 non-null   object 
 15  Math_Score                  1000 non-null   int64  
 16  Science_Score               1000 non-null   int64  
 17  Language_Score              1000 non-null   int64  
 18  History_Score               1000 non-null   int64  
 19  Average_Score               1000 non-null   int64  
 20  Grade                       1000 non-null   object 
dtypes: float64(4), int64(6), object(11)
memory usage: 164.2+ KB
```

```
df.describe()
```

	Age	Study_Hours_Per_Day	Sleep_Hours_Per_Day	Attendance_Rate
count	1000.000000	1000.000000	1000.000000	1000.000000
mean	21.453000	4.286350	7.048220	78.418000
std	2.293698	2.621616	1.242238	13.599243
min	18.000000	0.500000	2.500000	40.000000
25%	19.000000	2.387500	6.210000	69.275000
50%	21.000000	3.805000	7.080000	80.950000

75%	23.000000	5.815000	7.900000	89.100000
max	25.000000	15.500000	11.000000	99.800000

```
df.shape
```

```
(1000, 21)
```

```
df.columns
```

```
Index(['Name', 'Age', 'Gender', 'Major', 'Living_Area', 'Parent_Education',
       'Scholarship_Status', 'Part_Time_Job', 'Extracurricular_Activities',
       'Internet_Access', 'Study_Hours_Per_Day', 'Sleep_Hours_Per_Day',
       'Attendance_Rate', 'Stress_Level', 'Exam_Proximity', 'Math_Score',
       'Science_Score', 'Language_Score', 'History_Score', 'Average_Score',
       'Grade'],
      dtype='object')
```

5. Inquiry Question: Write a clear, specific question that you want to answer using your dataset by creating a text cell below and **write the question in bold please.**

For example:

- "How has the average temperature changed in major cities over the past decade?"
- "What factors are most strongly correlated with housing prices?"
- "Is there a relationship between education level and income?"

Is there any correlation between a student's major and their hours of sleep or stress level?

Part 2: Data Visualization and Analysis

TODO Tasks:

1. Data Cleaning (if necessary): Check for and handle any missing values, duplicates, or data quality issues.

NOTE: This could be a daunting task even for seasoned data scientists. My suggestion is to use a dataset that is already clean if you can for the purpose of completing this assignment in a week.

completing this assignment in a week.

Examples:

- Checking for missing values: df.isnull().sum()
- Dropping rows with any missing values: df.dropna(inplace=True)
- Dropping columns with all missing values: df.dropna(axis=1, how='all', inplace=True)
- Removing duplicate row: df.drop_duplicates(inplace=True)

2. Create Visualizations: Create at least 2 different types of visualizations that help answer your research question. Choose from:

- Bar charts
- Line plots
- Scatter plots
- Histograms
- Box plots
- Heatmaps
- Any other appropriate visualization

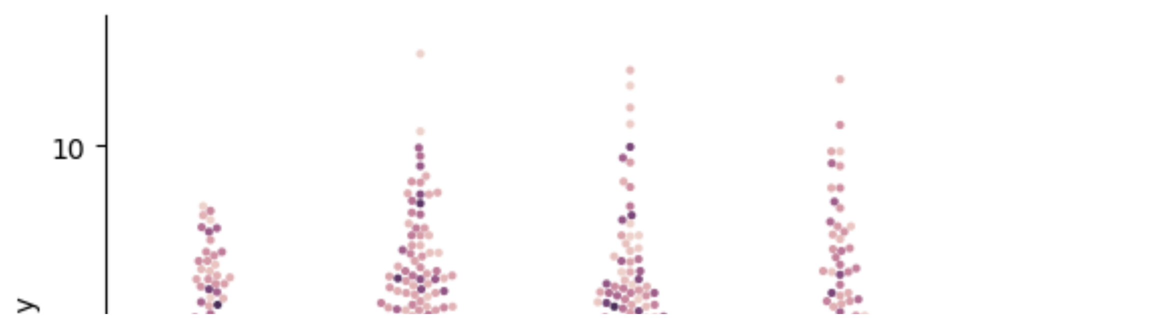
3. Customize Your Plots: Make sure your visualizations are clear and informative by:

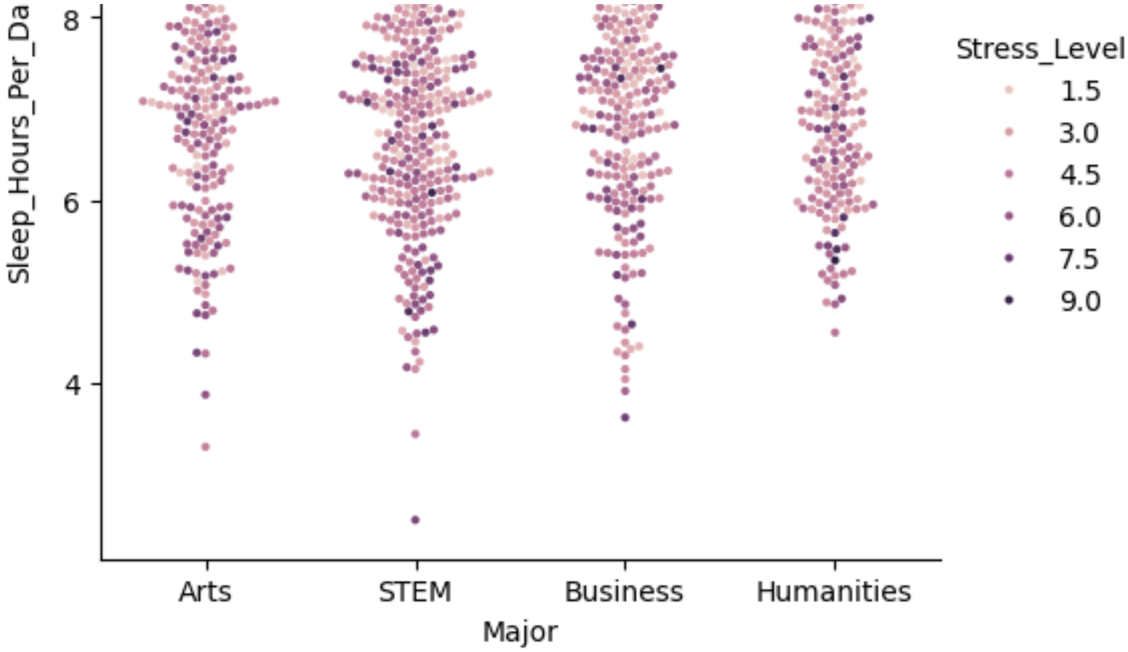
- Adding appropriate titles
- Labeling axes
- Using colors effectively
- Adding legends when necessary

Please create code cells below this to create your visualizations.

```
sns.catplot(data=df, x="Major", y="Sleep_Hours_Per_Day", hue="Stress_Level",
#swarm is used here to also see congregation on individual values - stress_l
```

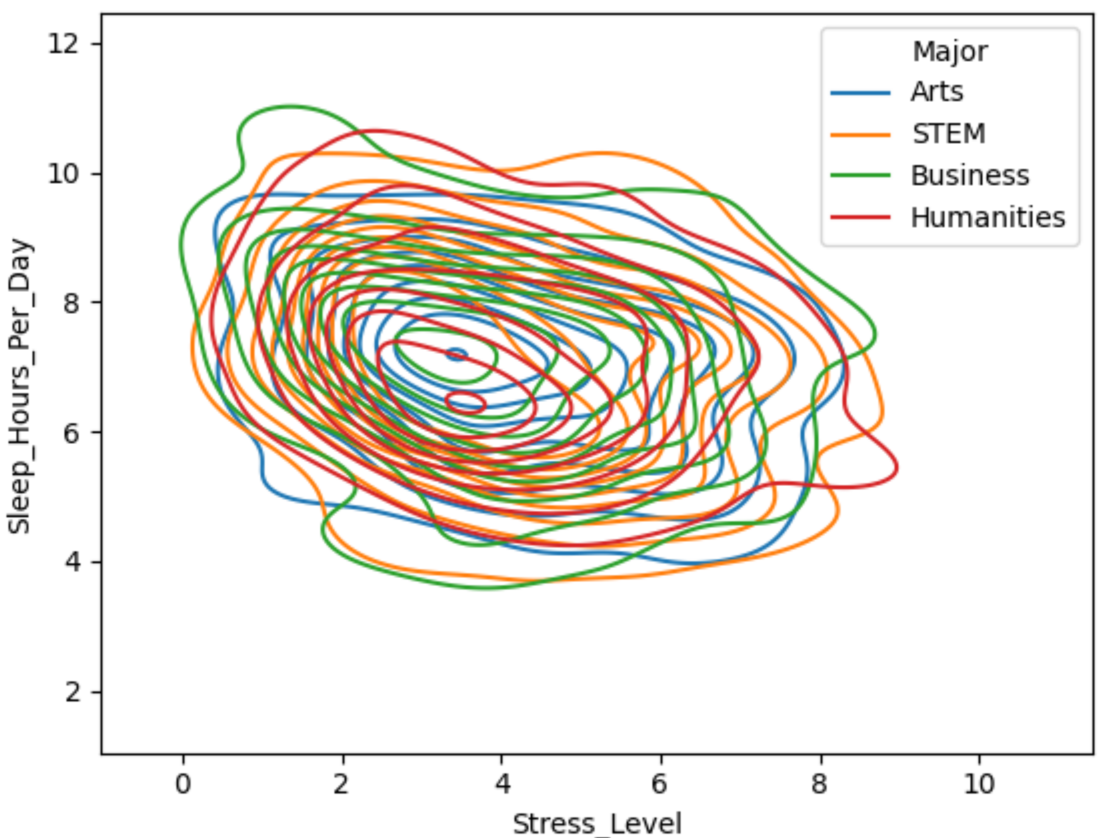
```
<seaborn.axisgrid.FacetGrid at 0x7b6f20bd7350>
```






```
sns.kdeplot(data=df, x="Stress_Level", y="Sleep_Hours_Per_Day", hue="Major")
#this one is also used to see concentration - more wide-view now - slight pe
```

```
<Axes: xlabel='Stress_Level', ylabel='Sleep_Hours_Per_Day'>
```



4. Analysis: Write 1-2 paragraphs discussing:

- Whether your research question can be answered with the available data

- What insights you discovered through your visualizations
- Any unexpected or interesting findings
- Limitations of your analysis
- Suggestions for further investigation

Please create a text cell below this cell to write your analysis.

The data I believe is able to answer the research question: no, there is no significant correlation between major and stress level or major and hours of sleep. The findings are fairly minor here: art and humanities majors have the most congregated data by a slight margin, whereas business and stem majors have more outliers in regards to hours of sleep and stress level. For the art and humanities majors, art majors trend to be slightly lower in hours of sleep (which is one of the few things I was expecting), and humanities higher. Based on the kdeplot, business majors trend slightly towards having the lowest stress level and getting the most sleep, but again - slightly - and so this should not be taken as definitive - these findings could be due to a limited dataset of only 1,000 students. A larger dataset may be necessary for a fuller analysis.

Submission Requirements

NOTE: If you use a local dataset, you MUST submit the dataset along with your notebook to get credit.

- Download your notebook as CS110_Project6.ipynb
- Ensure all TODO tasks are completed
- Make sure your code runs without errors
- Include clear comments explaining your work
- Submit the notebook file

Grading Criteria (5 points)

- Dataset Selection: Appropriate choice of dataset with sufficient complexity (0.5 point)
- Inquiry Question: Clear, specific, and answerable with the chosen data (1 point)

- Data Exploration: Proper use of pandas functions for initial analysis (0.5 point)
- Visualizations: At least 2 clear, well-formatted, and relevant plots (1 point)
- Analysis: Thoughtful interpretation of findings and answer to research question (1 point)
- Code Quality: Clean, commented, and well-organized code (1 point)

Remember: The goal is not just to create visualizations, but to use them to gain meaningful insights about your data. Think like a data scientist and let your curiosity guide your exploration!